

Firmware Installation Guide

Complete step-by-step instructions to install, configure, and flash both ESP32 boards for the ROBOTO glove-controlled robotic arm.

Prerequisites

Hardware Required

- 2× ESP32 WROOM-32 development boards (Glove + Arm)
 - 2× USB Micro-B cables (**data cables — not charge-only**)
 - A Windows / macOS / Linux computer
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Step 1 — Install Arduino IDE

1. Go to <https://www.arduino.cc/en/software>
 2. Download **Arduino IDE 2.x** (recommended) for your OS
 3. Run the installer and follow the prompts
 4. Launch Arduino IDE
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Step 2 — Install ESP32 Board Package

The Arduino IDE does not support ESP32 by default. You need to add it manually.

1. Open Arduino IDE
2. Go to **File** → **Preferences**
3. Find the field "**Additional boards manager URLs**"
4. Paste this URL into the field:

```
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
```

5. Click **OK**
6. Go to **Tools** → **Board** → **Boards Manager**
7. Search for esp32
8. Find "**esp32 by Espressif Systems**"
9. Click **Install** (version 2.0.x or later)
10. Wait for installation to complete (may take a few minutes)

Step 3 — Install Required Libraries

Go to **Sketch** → **Include Library** → **Manage Libraries** and install each of the following:

Library Name	Author	How to Search
Adafruit BNO055	Adafruit	Search BNO055
Adafruit Unified Sensor	Adafruit	Search Adafruit Unified Sensor
Adafruit PWM Servo Driver	Adafruit	Search PCA9685

Important: When installing Adafruit BNO055, Arduino IDE will ask: *"Install missing dependencies?"* — Click **"Install All"** This automatically installs Adafruit Unified Sensor as well.

Step 4 — Download the Firmware

Clone or download the repository:

```
# Option A — Git clone
git clone https://github.com/YOUR_USERNAME/robotic-glove-arm.git

# Option B — Download ZIP
# Go to your GitHub repo → Code → Download ZIP → Extract
```

Your folder structure should look like this after extracting:

```
robotic-glove-arm/
├── firmware/
│   ├── arm_esp32/
│   │   └── arm_esp32.ino
│   ├── glove_esp32/
│   │   └── glove_esp32.ino
│   └── shared/
│       └── config.h
```

Step 5 — Configure Board Settings

Every time you open a sketch, set these board settings in Arduino IDE before flashing:

Go to **Tools** menu and set:

Setting	Value
Board	ESP32 Dev Module
Upload Speed	921600
CPU Frequency	240MHz (WiFi/BT)

Setting	Value
Flash Frequency	80MHz
Flash Mode	QIO
Flash Size	4MB (32Mb)
Partition Scheme	Default 4MB with spiffs
Core Debug Level	None
PSRAM	Disabled

Step 6 — Flash the ARM ESP32 First

The arm must be flashed first so you can get its MAC address for the glove firmware.

6.1 — Open the sketch

1. In Arduino IDE go to **File** → **Open**
2. Navigate to `firmware/arm_esp32/arm_esp32.ino`
3. Click **Open**

6.2 — Select the correct port

1. Plug the **Arm ESP32** into your computer via USB
2. Go to **Tools** → **Port**
3. Select the port that appeared after plugging in:
 - Windows: COM3, COM4, COM5 etc.
 - macOS: `/dev/cu.usbserial-XXXX`
 - Linux: `/dev/ttyUSB0` or `/dev/ttyACM0`

Port not showing up? Install the CP2102 USB

driver: <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>

6.3 — Upload the firmware

1. Click the **Upload button** (→ arrow) in Arduino IDE
2. Wait for `Compiling sketch...` then `Uploading...`
3. You will see:

```
Writing at 0x00010000... (100 %)
Leaving...
Hard resetting via RTS pin...
```

4. Upload is complete

6.4 — Get the Arm MAC address

1. Go to **Tools** → **Serial Monitor**
2. Set baud rate to **115200** (bottom right dropdown)
3. Press the **EN/RST button** on the ESP32
4. Look for this line in the output:

```
[NET] Arm MAC: AA:BB:CC:DD:EE:FF
```

5. **Copy this MAC address** — you will need it in the next step
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Step 7 — Set the ARM MAC in Glove Firmware

1. Open `firmware/shared/config.h` in any text editor
2. Find this line near the top of Section 5:

```
#define ARM_MAC { 0xAA, 0xBB, 0xCC, 0xDD, 0xEE, 0xFF } // ← REPLACE THIS
```

3. Replace the placeholder bytes with your actual MAC address

Example: If your arm MAC is `3C:71:BF:4A:2D:08` change it to:

```
#define ARM_MAC { 0x3C, 0x71, 0xBF, 0x4A, 0x2D, 0x08 }
```

4. Save the file
-

Step 8 — Flash the GLOVE ESP32

8.1 — Open the sketch

1. Go to **File** → **Open**
2. Navigate to `firmware/glove_esp32/glove_esp32.ino`
3. Click **Open**

8.2 — Select the correct port

1. **Unplug** the Arm ESP32
2. **Plug in** the Glove ESP32
3. Go to **Tools** → **Port** and select the new port

8.3 — Upload the firmware

1. Click the **Upload button**
2. Wait for upload to complete (same process as Step 6.3)

8.4 — Verify glove is running

1. Open **Serial Monitor** at **115200 baud**
2. Press the **EN/RST button** on the Glove ESP32
3. You should see:

```
[GLOVE] Booting...
[OK] BN0055 ready
[CAL] Offsets loaded from NVS
[NET] Glove MAC: XX:XX:XX:XX:XX:XX
[OK] ESP-NOW ready → Arm MAC set
[GLOVE] Running – hold BOOT 3 s to calibrate
```

Step 9 — Verify Full System Communication

1. Power on the **Arm ESP32** (plug into USB or external supply)
2. Open Serial Monitor on the **Arm** board at 115200 baud
3. Power on the **Glove ESP32**
4. Move the glove — you should see live data on the arm Serial Monitor:

```
B:135 S:115 E:120 WP:135 WR:135 G:30 | RX R:0.0 P:0.0 Y:0.0
B:138 S:112 E:118 WP:140 WR:130 G:30 | RX R:5.2 P:-3.1 Y:2.8
```

5. The arm status LED (GPIO 2) should be **solid ON** when receiving packets

LED blinking? No signal is being received — check MAC address in config.h and re-flash glove firmware.

Step 10 — Run Calibration

Glove calibration (always run first)

1. Hold the **BOOT button on the Glove ESP32** for **3 seconds**
2. Open Serial Monitor on the glove — follow the prompts:

```
===== CALIBRATION MODE =====
Keep glove FLAT & STILL (neutral pose)...
Starting in 5 s
```

3. Step through all three stages:

Stage	Action	Duration
Neutral pose	Hold hand flat and still	5 s countdown then 1 s sample
Flex OPEN	Open hand fully and hold	3 s
Flex CLOSE	Close hand into a fist and hold	3 s

1. Offsets are saved automatically to NVS (survive power cycles)
2. LED blinks 6 times = calibration saved successfully

Arm trim calibration (run after glove calibration)

1. Hold the **BOOT button on the Arm ESP32** for **3 seconds**
2. Open Serial Monitor on the arm — you will see:

```

===== ARM CALIBRATION MODE =====
Commands:
  t<ch> <deg> - set trim for channel (e.g. t0 -5.0)
  s<r|p|y> <f> - set scale (e.g. sr 0.8)
  h           - send arm to HOME
  q           - save & quit

```

3. Use the commands to fine-tune each servo:

Command	Example	Effect
t<ch> <deg>	t0 -5.0	Shift base servo -5° from computed angle
t<ch> <deg>	t3 3.5	Shift wrist pitch +3.5°
s<axis> <val>	sr 0.75	Reduce roll sensitivity to 75%
s<axis> <val>	sy 1.2	Increase yaw sensitivity to 120%
h	h	Move all servos to HOME position
q	q	Save all trims to NVS and exit

1. Type q when done — settings are saved permanently

Troubleshooting

Problem	Likely Cause	Fix
Port not visible	Missing USB driver	Install CP2102 driver from Silicon Labs
BN0055 not found	Wrong wiring or 5V on VIN	Check SDA=21, SCL=22, VIN=3.3V only
Upload fails / timeout	Wrong port selected	Unplug, replug, re-select port
Upload fails — stuck at Connecting...	Boot mode issue	Hold BOOT button while clicking Upload, release when Uploading... appears
Arm LED blinking	No ESP-NOW signal	Check MAC address in config.h, re-flash glove
Servos not moving	Servo power rail issue	Check V+ screw terminal on

Problem	Likely Cause	Fix
		PCA9685, verify external supply GND tied to ESP32 GND
Jittery movement	Power supply too weak	Use a dedicated 5V/3A BEC for servos
Gripper not triggering	Flex calibration wrong	Re-run glove calibration (BOOT 3 s)
Wrong servo direction	Map() range inverted	Swap minDeg and maxDeg in SERVO_CFG[] inside config.h

Re-flashing After Changes

Whenever you edit `config.h`:

- **Both boards must be re-flashed** since `config.h` is shared
 - Always flash **Arm first**, then **Glove**
 - Re-run glove calibration after any re-flash to restore NVS offsets
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Serial Monitor Quick Reference

Board	Baud Rate	What you see
Glove	115200	Roll / Pitch / Yaw / Flex raw / Grip state
Arm	115200	6 servo angles + received R/P/Y values
